Om Patel

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Education

University of Toronto – BASc in Electrical Engineering + PEY Co-op **Expected May 2028**

- **Relevant Coursework**: Signals and Systems, Digital Systems, Analog & Digital Electronics, Hardware Design & Communication, Computer Networks & Hardware, Fundamentals of Deep Learning, Control Systems

Experience

Quality & Automation Engineer Intern | *Relay Financial (Fintech Startup)* **May 2026 – Present**

- Built core banking test suite to 50+ maintained tests by writing 15+ end-to-end tests (API, UI, database) for API integrations (Air Wallex, Nium International Wires) using Playwright and Bruno
- Configured CI/CD pipeline for mobile deployments using GitHub Actions and Detox, establishing automated smoke testing infrastructure for the mobile engineering team, enabling faster and safer release cycles
- Developed in-house simulation API templates for specific account/organization/user creation, reducing smoke test setup time and enabling API-driven test execution instead of manual account creation

Analyst Software Tester Intern | *FNZ (Fintech Private)* **May 2025 – Aug 2025**

- Reduced pending retest backlog by 50%+ and unblocked team specific development by identifying 30+ platform bugs and validating 75+ fixes across staging and user-test environments using SQL, RabbitMQ, API logs, and Postman
- Improved all-round automated regression test accuracy by 18% by debugging 45+ Cypress test cases with JavaScript
- Revamped Client Onboarding QA by converting manual functionality coverage into an E2E test plan and automated Cypress test suite covering 10+ workflows, happy paths, edge cases, and core platform functionality
- Delivered signed-off requirements for the FNZ-BMO Fund Classification workflow by mapping 2 market-data XML feeds (Bloomberg, Morningstar) to database fields and authoring 6+ user stories for downstream fund calculations

Data Analyst & Administrative Assistant | *University of Toronto Engineering Career Centre* **May 2024 – Aug 2024**

- Presented PEY Co-op insights and concrete improvement steps to 15+ staff by analyzing 1,200+ survey responses with regards to student interests, industry preferences and service quality in Excel
- Produced Annual and Seasonal Recruitment Reports by analyzing 2,000+ Excel student records to identify hiring trends

Projects

CNN Land-Use Classification Model | *Python, PyTorch, torchvision, scikit-learn, pandas* **Report** | **Apr 2025**

- Achieved 96.9% test accuracy on 10-class EuroSAT land-use classification by training a 2.39M-parameter PyTorch CNN with 6 convolutional layers, batch normalization, dropout, and Adam optimization, outperforming a 50.7% HoG baseline
- Built a PyTorch training pipeline with torchvision, pandas, and scikit-learn, scaling EuroSAT training data from 18.9k to 94.5k images through 5x augmentation while supporting stratified 70/15/15 splits, GPU training, and checkpointing

Bread Crumb Trail | MakeUofT 2026 HW Hackathon, Winner | *Arduino UNO Q, Python, C++* **DevPost** | **Feb 2026**

- Engineered a navigation and SOS system on Arduino UNO Q (AppLab), integrating GNSS/GPS (I2C), SSD1306 OLED (I2C), and a GSM/LTE cellular module (UART) to provide live coordinates and breadcrumbs to Streamlit dashboard
- Implemented cloud integrations: Twilio API for SOS SMS (compressed trail summary), Google Maps API for route/waypoint visualization, and Gemini API for interactive emergency guidance and breadcrumb-based path prediction
- Awarded the Qualcomm Sponsor Prize, Best Use of Arduino UNO Q (out of 260+ submissions)

Radio Power Amplifier and Low-Pass Filter PCB | *Altium, LTSpice, Soldering* **Github** | **Jan 2025 - May 2025**

- Constructed a Class E power amplifier and 7-stage maximally flat low-pass filter, obtaining 2.7W output power across antenna load (30dB gain) and 1.69% total harmonic distortion, meeting requirements for radio integration
- Designed a 2-layer PCB in Altium Designer for 16MHz and 60V operation, achieving 76% power efficiency
- Executed automated hardware unit testing using Python to interface with oscilloscope, DMM, and waveform generator

B.O.B | MakeUofT 2025 HW Hackathon, 2x Winner | *Arduino UNO R3, Sensors* **DevPost** | **Feb 2025**

- Developed a smart recycling bin that automatically opened/closed based on available capacity (ultrasonic, IR sensors) and determined recyclability using Gemini AI and OpenFoodFacts APIs
- Awarded Best Hack for Sustainability & Green Tech (out of 20+ teams) and Best .Tech Domain Name (out of 60+ teams)

Skills

Languages: Python, TypeScript, JavaScript, SQL, C, C++, RV32 Assembly, Verilog
Tools: Cursor, Docker, AWS, Datadog, GitHub Actions/Workflows, JIRA, Postman/Insomnia/Bruno, Postgres/MySQL
Data & ML: PyTorch, Pandas, scikit-learn, torchvision, Matplotlib, NumPy
Frameworks & Testing: React.js, Express.js, Node.js, Playwright, Cypress